

Definiciones

La función de error está dada por:

$$E(W) = \frac{1}{2} \sum_{\mu} \sum_i (\zeta_i^{\mu} - O_i^{\mu})^2 \quad (1)$$

donde la salida de la red para el ejemplo μ y la unidad i es:

$$O_i^{\mu} = g(h_i^{\mu}) \quad (2)$$

y

$$h_i^{\mu} = \sum_j W_{ij} V_j^{\mu} \quad (3)$$

Objetivo

Queremos calcular ΔW_{ij} :

$$\Delta W_{ij} = -\eta \frac{\partial E}{\partial W_{ij}} \quad (4)$$

Derivación

Partimos de la regla de la cadena en forma compacta:

$$\frac{\partial E}{\partial W_{ij}} = \sum_{\mu} \frac{\partial E}{\partial h_i^{\mu}} \frac{\partial h_i^{\mu}}{\partial W_{ij}} \quad (5)$$

Paso 1: Derivada del error respecto de h_i^{μ}

Sabemos que:

$$E = \frac{1}{2} \sum_{\mu, i} (\zeta_i^{\mu} - O_i^{\mu})^2 \quad \text{con} \quad O_i^{\mu} = g(h_i^{\mu}) \quad (6)$$

Entonces:

$$\frac{\partial E}{\partial h_i^{\mu}} = \frac{\partial E}{\partial O_i^{\mu}} \cdot \frac{\partial O_i^{\mu}}{\partial h_i^{\mu}} \quad (7)$$

$$\frac{\partial E}{\partial h_i^{\mu}} = -(\zeta_i^{\mu} - O_i^{\mu}) g'(h_i^{\mu}) \quad (8)$$

Paso 2: Derivada de h_i^μ

$$h_i^\mu = \sum_j W_{ij} V_j^\mu \quad (9)$$

$$\frac{\partial h_i^\mu}{\partial W_{ij}} = V_j^\mu \quad (10)$$

Resultado

Reemplazando:

$$\frac{\partial E}{\partial W_{ij}} = \sum_\mu [-(\zeta_i^\mu - O_i^\mu) g'(h_i^\mu)] V_j^\mu \quad (11)$$

$$\frac{\partial E}{\partial W_{ij}} = - \sum_\mu (\zeta_i^\mu - O_i^\mu) g'(h_i^\mu) V_j^\mu \quad (12)$$

Regla de actualización

$$\Delta W_{ij} = -\eta \frac{\partial E}{\partial W_{ij}} \quad (13)$$

$$\Delta W_{ij} = \eta \sum_\mu (\zeta_i^\mu - O_i^\mu) g'(h_i^\mu) V_j^\mu \quad (14)$$

Definiendo:

$$\delta_i^\mu = (\zeta_i^\mu - O_i^\mu) g'(h_i^\mu) \quad (15)$$

$$\Delta W_{ij} = \eta \sum_\mu \delta_i^\mu V_j^\mu \quad (16)$$

Derivación para Δw_{jk}

Queremos calcular:

$$\frac{\partial E}{\partial w_{jk}} \quad (17)$$

Aplicamos regla de la cadena considerando la dependencia a través de la capa oculta:

$$\frac{\partial E}{\partial w_{jk}} = \sum_\mu \frac{\partial E}{\partial V_j^\mu} \frac{\partial V_j^\mu}{\partial h_j^\mu} \frac{\partial h_j^\mu}{\partial w_{jk}} \quad (18)$$

Paso 1: Derivada respecto de V_j^μ

La variable V_j^μ afecta a todas las salidas:

$$\frac{\partial E}{\partial V_j^\mu} = \sum_i \frac{\partial E}{\partial h_i^\mu} \frac{\partial h_i^\mu}{\partial V_j^\mu} \quad (19)$$

Sabemos que:

$$h_i^\mu = \sum_j W_{ij} V_j^\mu \Rightarrow \frac{\partial h_i^\mu}{\partial V_j^\mu} = W_{ij} \quad (20)$$

Entonces:

$$\frac{\partial E}{\partial V_j^\mu} = \sum_i \frac{\partial E}{\partial h_i^\mu} W_{ij} \quad (21)$$

Paso 2: Derivadas restantes

$$V_j^\mu = g(h_j^\mu) \Rightarrow \frac{\partial V_j^\mu}{\partial h_j^\mu} = g'(h_j^\mu) \quad (22)$$

$$h_j^\mu = \sum_k w_{jk} \xi_k^\mu \Rightarrow \frac{\partial h_j^\mu}{\partial w_{jk}} = \xi_k^\mu \quad (23)$$

Resultado

Reemplazando todo:

$$\frac{\partial E}{\partial w_{jk}} = \sum_\mu \left[\left(\sum_i \frac{\partial E}{\partial h_i^\mu} W_{ij} \right) g'(h_j^\mu) \xi_k^\mu \right] \quad (24)$$

Usando que:

$$\frac{\partial E}{\partial h_i^\mu} = -\delta_i^\mu \quad (25)$$

obtenemos:

$$\frac{\partial E}{\partial w_{jk}} = - \sum_\mu \left[\left(\sum_i \delta_i^\mu W_{ij} \right) g'(h_j^\mu) \xi_k^\mu \right] \quad (26)$$

Regla de actualización

$$\Delta w_{jk} = -\eta \frac{\partial E}{\partial w_{jk}} \quad (27)$$

$$\Delta w_{jk} = \eta \sum_{\mu} \left[\left(\sum_i \delta_i^{\mu} W_{ij} \right) g'(h_j^{\mu}) \xi_k^{\mu} \right] \quad (28)$$

Definiendo:

$$\delta_j^{\mu} = g'(h_j^{\mu}) \sum_i W_{ij} \delta_i^{\mu} \quad (29)$$

llegamos a la forma final:

$$\Delta w_{jk} = \eta \sum_{\mu} \delta_j^{\mu} \xi_k^{\mu} \quad (30)$$